

Question 1

The domain of this model is football match outcome prediction, focusing on factors that influence match outcomes. Football matches are difficult to predict due to the fast-paced environment and the continuous flow of play, where shifts in momentum between teams frequently occur. The decision problem addresses the factors that influence the probability of winning, losing or drawing games. An Influence Diagram enables decision-making under uncertainty and provides a suitable structure to model complex relationships between key factors making it suitable for this problem given the unpredictable nature of football matches.

I will be using human knowledge to construct my model. My Bayesian network model will consist of the following 8 nodes :

- Match Outcome: Final match result
- Team Attack: Team attacking capability
- Team Defence: Team defensive capability
- Opponent Attack: Opponent attacking capability
- Opponent Defence: Opponent defensive capability
- Shots on Target (For): Team attacking performance
- Shots on Target (Against): Team defensive performance
- Home Advantage: Home or away status

The different relationships between the nodes in my Bayesian network model are used to reflect the real-world situations associated with football games. Team Attack and Opponent Defence influence the number of shots on target for a team, as stronger attacking teams are likely to create clear goal-scoring opportunities against weak defences. Constant attacking pressure improves the chances of scoring goals and increases the likelihood of winning. General measures of offensive effectiveness such as shots on goal have been proposed as good indicators of successful match results (Sarmento et al., 2018). On the other hand, Team Defence and Opponent Attack influence the number of shots on target against a team, as weaker defensive units are likely to be vulnerable to threats from strong attacking players. Conceding a higher number of shots is strongly associated with conceding more goals, as increased defensive pressure creates more goal-scoring opportunities (Prieto-González et al., 2025). Furthermore, home advantage directly influences match outcomes as the crowd support can cause opposition players to crumble under pressure while positively impacting player performance and contributing to a measurable home advantage effect (Ponzo and Scoppa, 2018).

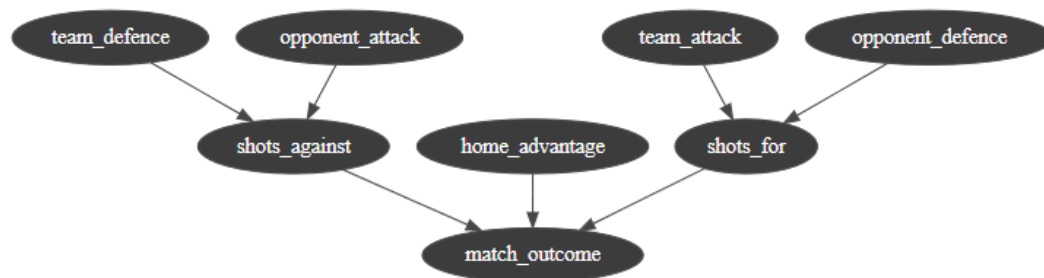
References

Sarmento, H., Figueiredo, A., Lago-Peñas, C., Milanovic, Z., Barbosa, A., Tadeu, P. and Bradley, P.S. (2018). Influence of Tactical and Situational Variables on Offensive Sequences During Elite Football Matches. *Journal of Strength and Conditioning Research*, 32(8), pp.2331–2339. doi:<https://doi.org/10.1519/jsc.0000000000002147>.

Prieto-González P, Martín V, Sal-de-Rellán A. Impact of defensive team variables on goals conceded in the first division of the Spanish soccer league: a 10-year study. *Biology of Sport*. 2025;42(3):17-28. doi:10.5114/biol sport.2025.145914.

Ponzo, Michela and Scoppa, Vincenzo, Does the Home Advantage Depend on Crowd Support? Evidence from Same-Stadium Derbies. IZA Discussion Paper No. 8105, Available at SSRN: <https://ssrn.com/abstract=2426859>

Question 2



To construct my Bayesian Network model, all nodes were declared as labeled variables and arcs which were representative of the relationships between the nodes were added. Once the nodes and arcs had been created, my Bayesian Network model could be declared and visualised.

The model was constructed from my own knowledge of football concepts where key variables affecting match outcomes were identified and linked through relationships. Developing my model involved initialising the key variables influencing the results of football matches and establishing key relationships between these variables. For example, Team defence strength and opponent attack strength variables were defined as parent nodes of shots on target (against) while Team Attack strength and opponent defence strength were defined as parent nodes of shots on target (for). These shots on target variables along with home advantage were declared as parent nodes of the match outcome node which represents the final decision variable.

This knowledge-based approach suits my problem because relationships representative of real-world problems in football can be modelled simply without the need of any extra steps such as data preprocessing. With this approach, key variables and relationships affecting match outcomes can be defined without any additional complexity.

Question 3

home_advantage	
0	1
0.5000	0.5000

*#For Home Advantage:
0 represents 'Away'
1 represents 'Home'*

This CPT represents the probability of a team playing at home or away. As a root node, it has no parent nodes due to the match venue being independent of other factors. This CPT was manually constructed and had a low complexity due to having no parents.

		shots_for		
opponent_defence	team_attack	0	1	2
0	0	0.3000	0.4000	0.3000
	1	0.2500	0.3000	0.4500
	2	0.1000	0.2500	0.6500
1	0	0.4500	0.3000	0.2500
	1	0.3000	0.4000	0.3000
	2	0.2000	0.3000	0.5000
2	0	0.6500	0.2500	0.1000
	1	0.5000	0.3000	0.2000
	2	0.2500	0.5000	0.2500

#For Shots on Target(For):
0 represents 'Low'
1 represents 'Medium'
2 represents 'High'

This CPT represents the probability of a team getting shots on target based on its two parent nodes . A stronger attack and weaker opponent defence increases the likelihood of the team attempting more shots on target.This CPT was manually constructed and had moderate complexity due to having two parents.

		shots_against		
opponent_attack	team_defence	0	1	2
0	0	0.3000	0.4000	0.3000
	1	0.4500	0.3000	0.2500
	2	0.6500	0.2500	0.1000
1	0	0.2500	0.3000	0.4500
	1	0.3000	0.4000	0.3000
	2	0.5000	0.3000	0.2000
2	0	0.1000	0.2500	0.6500
	1	0.2500	0.3000	0.4500
	2	0.2500	0.5000	0.2500

#For Shots on Target(Against):
0 represents 'Low'
1 represents 'Medium'
2 represents 'High'

This CPT represents the probability of a team conceding shots on target based on its two parent nodes . A weaker defence and stronger opponent attack increases the likelihood of the team facing more shots on target.This CPT was manually constructed and had moderate complexity due to having two parents.

			match_outcome		
shots_against	home_advantage	shots_for	0	1	2
0	0	0	0.3500	0.4000	0.2500
		1	0.3000	0.3000	0.4000
		2	0.2000	0.3000	0.5000
	1	0	0.3000	0.4000	0.3000
		1	0.2000	0.3000	0.5000
		2	0.1000	0.2000	0.7000
1	0	0	0.5000	0.3000	0.2000
		1	0.3000	0.4000	0.3000
		2	0.3000	0.3000	0.4000
	1	0	0.4000	0.3500	0.2500
		1	0.2500	0.3500	0.4000
		2	0.2000	0.3000	0.5000
2	0	0	0.6000	0.3000	0.1000
		1	0.5000	0.3000	0.2000
		2	0.3500	0.3500	0.3000
	1	0	0.5000	0.3000	0.2000
		1	0.4000	0.3500	0.2500
		2	0.3000	0.3000	0.4000

#For Match Outcome:
0 represents a 'Loss'
1 represents a 'Draw'
2 represents a 'Win'

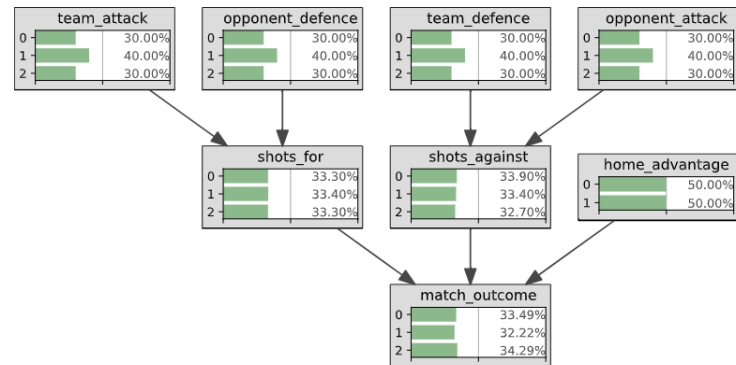
This CPT represents the probability of the match outcome based on its 3 parents.Having a home advantage , attempting more shots and conceding less shots on target increases the likelihood of a team winning the game. This CPT was manually constructed and had a high level of complexity due to having 3 parents.

Question 4

Prior Distribution :

match_outcome		
0	1	2
0.3349	0.3222	0.3429

$SP(match_outcome)$



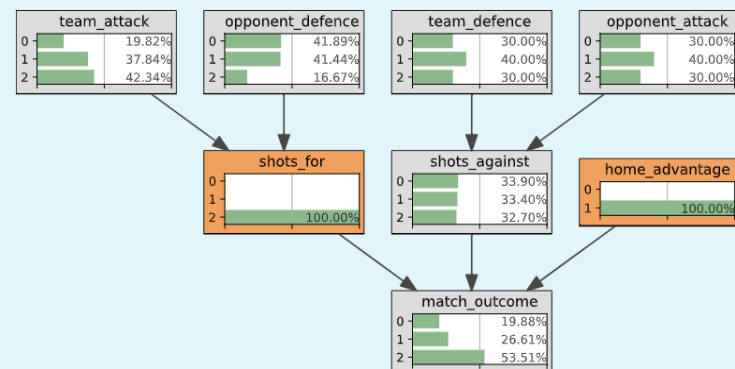
Inference in 1.16ms

Complete inference with NO evidence

Posterior Distribution

match_outcome		
0	1	2
0.1988	0.2661	0.5351

$SP(match_outcome|home_advantage = 1, shots_for = 2)$



Inference in 0.83ms

Complete inference with evidence = {(home_advantage:1, shots_for:2)}

Observational reasoning was carried out by introducing evidence that would affect the probability of the Match Outcome node. Initially , the prior distribution of the Match Outcome node was computed with no observed evidence. The posterior distribution was then computed once the observed evidence had been declared with the home advantage node being set to 1(home) and the shots for node being set to 2 (high). As expected , the probability of the team winning the match increased as generating a high number of shots on target and having a home advantage would lead to more goals being scored.

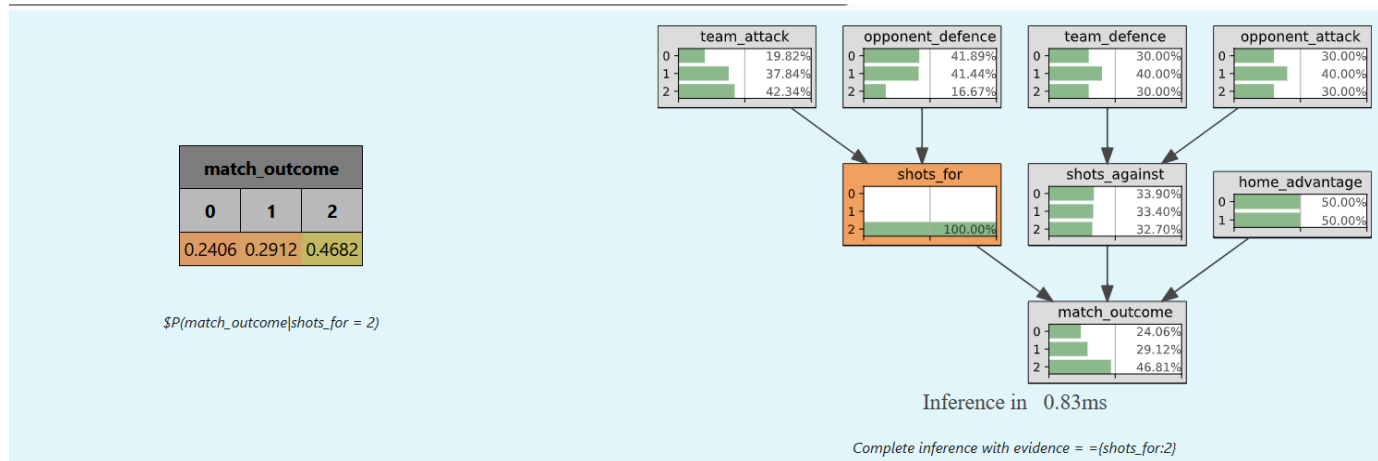
Interventional Reasoning :

$$P(\text{match_outcome} \mid \text{do}(\text{shots_for})) = \sum_{\text{home_advantage}, \text{opponent_attack}, \text{shots_against}, \text{team_defence}} P(\text{opponent_attack} \mid \text{shots_for})$$

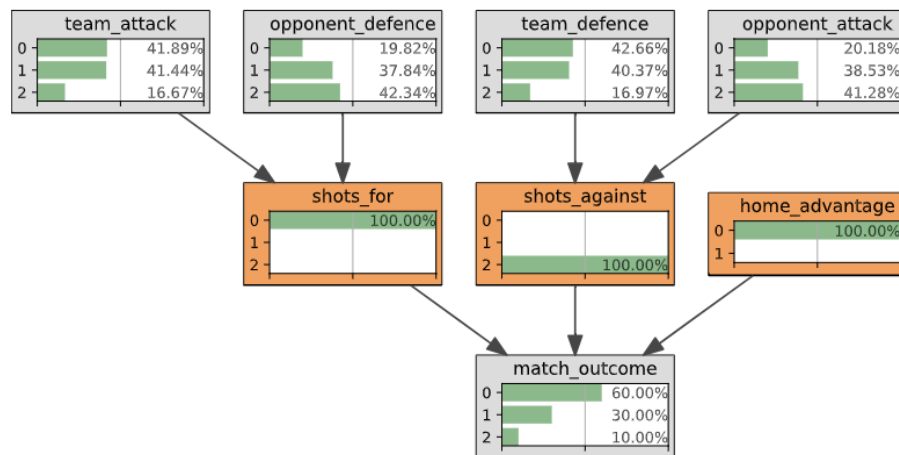
match_outcome		
0	1	2
0.2406	0.2912	0.4681

Impact

Comparison with observational case on same node :

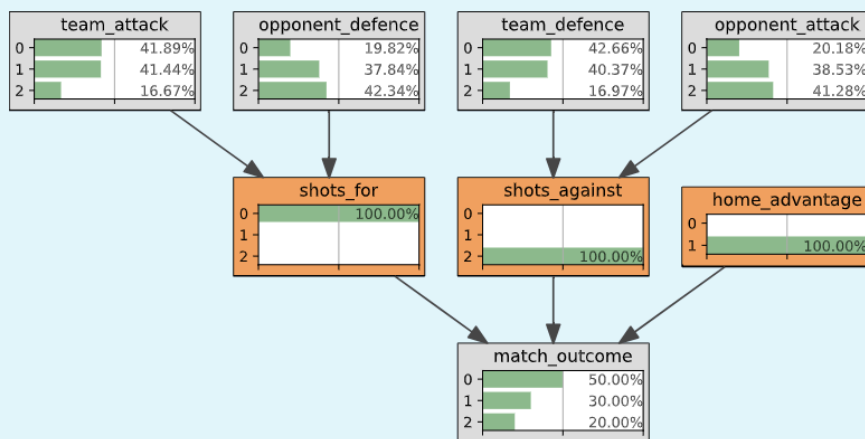


Interventional reasoning was carried out by applying the 'do -operator' to set the 'shots_for' node to 2 (high). The probability distribution of match outcomes was equivalent to the observational case suggesting there were no other external variables influencing this relationship. This suggests that there is a direct relationship between the 'shots_for' and 'match_outcome' nodes. Conditioning involves observing a variable and updating its probabilities based on its relationships with its parent nodes whereas intervention forces a value to a variable and does not take into account the parent nodes.



Inference in 2.72ms

real world

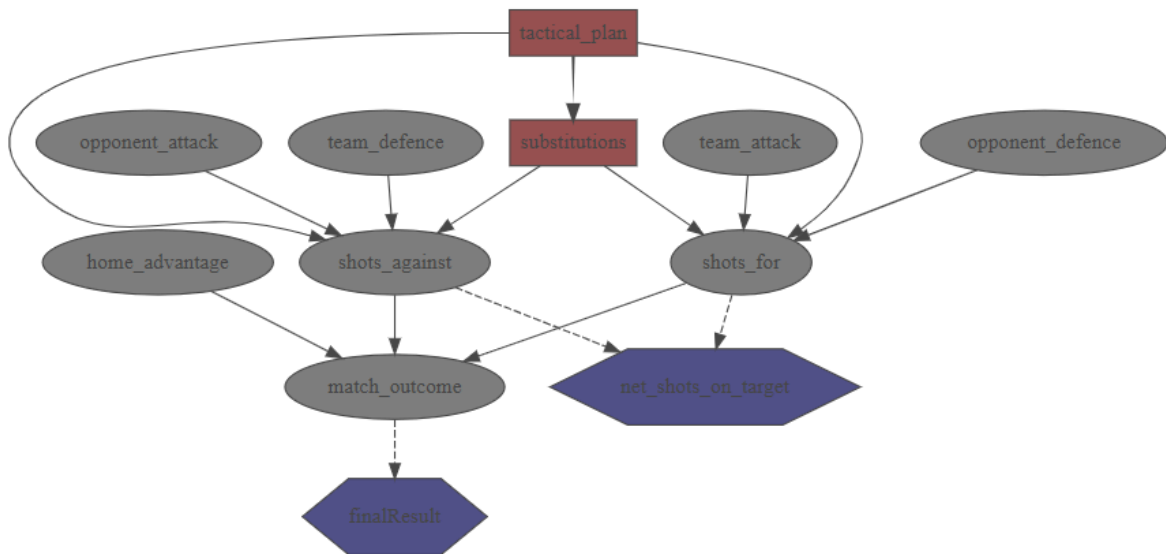


Inference in 1.05ms

hypothetical world

Counterfactual reasoning was carried out by comparing an observed situation to a hypothetical situation. With the observed situation, The team was playing an away match with a low amount of shots on target and a high amount of shots against. An alternative counterfactual model was then declared which then simulated the same scenario as the observed situation but with the team playing at home. As expected, the probability of the team losing decreased and the probability of the team winning increased demonstrating how home advantage can positively influence match outcomes.

Question 5



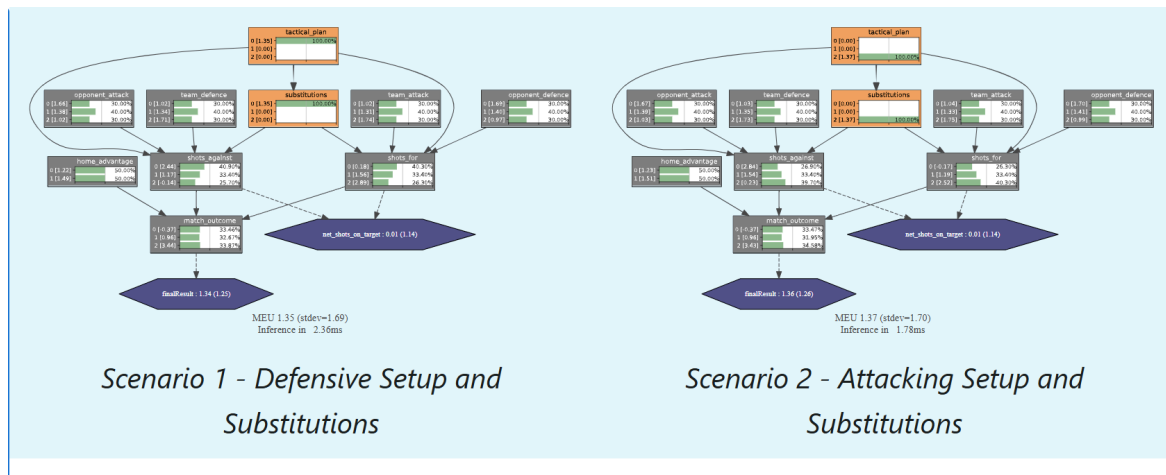
The decision node 'tactical_plan' represents the team's style of play(defensive , balanced or attacking),influencing the attacking and defensive performance.Team that have a good defence would choose to play a defensive style of football while teams with explosive attack would choose to play an attacking style of football.

The decision node 'substitutions' represents the in-game tactical changes that are made to improve the team's overall performance.Substitutions of players can either be defensive , balanced or attacking and are used to influence the results of matches by improving the attacking or defensive performance of the team.

The utility node 'finalResult' represents the most preferred outcomes of the match. The utility values were defined based on outcomes and a football point scoring system was used where 3 points would represent a win , 1 point would represent a draw and 0 points would represent a loss.

The utility node 'net_shots_on_target' represents the overall dominance of the game determined from calculating the difference between the shots on target for the team and shots on target against the team. The utility values were defined based on outcomes where positive values would represent desirable outcomes(good attacking performance) and negative values representing undesirable outcomes(weak defensive performance).

Question 6



One scenario was when the team had a defensive tactical plan with defensive substitutions, while another scenario was when the team had an attacking tactical plan with attacking substitutions. The attacking scenario computed a higher MEU (1.37) value than the defensive scenario (1.35) suggesting that an attacking approach to a game is more likely to result in better outcomes.

An attacking setup results in a higher probability of a team attempting more shots on target which would lead to more games being won as more goals would be scored. However, an attacking setup would also come at the cost of facing more shots on target as some defensive capability of the team would have to be sacrificed to facilitate an attacking approach. Despite this, the return of an attacking approach would outweigh the risks of a less defensive approach. Due to this reasoning, an attacking setup would positively influence the 'net_shots_on_target' and 'finalResult' utility nodes.

With the defensive scenario, a lower MEU value was obtained which can be explained by a defensive approach limiting the attacking force of a team leading to fewer shots on target. Despite defensive stability being a possibility, the attacking output is significantly reduced with this approach, lowering the likelihood of scoring goals and accomplishing higher utility outcomes such as winning the game.

tactical_plan		
0	1	2
0.0000	0.0000	1.0000

substitutions		
0	1	2
0.0000	0.0000	1.0000

MEU = 1.3689418000000002

Optimal Decision for Tactical Plan

Optimal Decision for Substitutions

Maximum Expected Utility (MEU)

The optimal decision was computed for both decision nodes and in both cases, an attacking tactical plan and substitutions was the output. With reference to the model's behaviour in the context of football matches, these outputs are appropriate because an attacking approach

to a football game would be more rewarding in terms of final match results compared to a defensive approach. Overall , these results are reasonable and realistic.

Question 7

One strength of my model is that it accurately models the relationships between the team attack and team defence strength to the shots on target for and against. These relationships make logical sense as a team that has a very strong attack is likely to attempt a higher number of shots on target against a team that has a weak defence.

One weakness of my model is that it considers shots on target for and against as the primary factors of determining match outcomes. In a football match, loads of different factors such as possession , crosses in the box and set pieces can influence the amount of goals scored and conceded by a team. My model does not take into account these factors.

To improve the accuracy of my model , additional factors affecting match outcome such as possession , crosses in the box and set pieces could be considered. Considering different factors that affect match outcomes would make the model more realistic because football results are dependent on a large number of different variables and this makes football matches incredibly difficult to predict.